

10.2 The Cholesky factorization

If the cross-product matrix is to be used at all, the normal equations should be solved by factoring it as a triangular matrix times its transpose. One factorization of the augmented cross-product matrix then delivers every standard regression quantity.

10.2.1 The algorithm

Chapter 1 proved that a positive definite matrix has a unique factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ lower triangular and positive on the diagonal (Theorem 1.7.4(d)). Numerical work usually writes it with the upper triangular factor $\mathbf{R} = \mathbf{L}^\top$.

Theorem 10.2.1. Cholesky factorization

Let $\mathbf{A}$ be a $p \times p$ positive definite matrix.

- (a) There is exactly one upper triangular $\mathbf{R}$ with positive diagonal entries such that $\mathbf{A} = \mathbf{R}^\top \mathbf{R}$.
- (b) Its rows are given, for $j = 1, \dots, p$ in turn, by

$$\begin{aligned} r_{jj} &= \left(a_{jj} - \sum_{k < j} r_{kj}^2 \right)^{1/2}, \\ r_{ji} &= \frac{1}{r_{jj}} \left(a_{ji} - \sum_{k < j} r_{kj} r_{ki} \right), \quad i > j, \end{aligned} \tag{10.2.1}$$

and every radicand in it is positive.

- (c) The computation takes $p^3/3 + O(p^2)$ flops.
- (d) If $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{X} = \mathbf{Q}\mathbf{R}_0$ is a QR factorization with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}_0$ upper triangular with positive diagonal, then $\mathbf{R}_0 = \mathbf{R}$.

Proof.

- (a) is Theorem 1.7.4(d) with $\mathbf{R} = \mathbf{L}^\top$.
- (b) Entry (j, i) , $i \geq j$, of $\mathbf{R}^\top \mathbf{R}$ is $\sum_{k \leq j} r_{kj} r_{ki}$, because $r_{kj} = 0$ for $k > j$. Setting it equal to a_{ji} and isolating the $k = j$ term gives the recursion Eq. 10.2.1. So the entries of the unique factor of (a) satisfy Eq. 10.2.1, and Eq. 10.2.1 determines row j from rows $1, \dots, j-1$. By induction on j , the rows computed by Eq. 10.2.1 are those of the factor in (a). In particular the j th radicand is $r_{jj}^2 > 0$.
- (c) Row j needs about $2(j-1)$ flops for each of its $p-j+1$ entries, so the total is $\sum_j 2(j-1)(p-j+1) = p^3/3 + O(p^2)$.
- (d) $\mathbf{X}^\top \mathbf{X} = \mathbf{R}_0^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{R}_0 = \mathbf{R}_0^\top \mathbf{R}_0$, and uniqueness in (a).

■

By part (d), the triangle of a QR factorization of $\mathbf{X}$ and the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$ are the same matrix (see also Exercise (B2)). The two routes of this chapter compute the same object, one from $\mathbf{X}$, the other from $\mathbf{X}^\top \mathbf{X}$ after the small singular values may already have been damaged (Proposition 10.1.3).

The algorithm itself is backward stable *for the factorization*. The computed $\hat{\mathbf{R}}$ satisfies $\hat{\mathbf{R}}^\top \hat{\mathbf{R}} = \mathbf{A} + \Delta \mathbf{A}$ with $|\Delta \mathbf{A}| \leq \gamma_{p+1} |\hat{\mathbf{R}}|^\top |\hat{\mathbf{R}}|$ entrywise, and it runs to completion whenever $\mathbf{A}$ is not too close to singular (Higham 2002, chapter 10), with no pivoting. The weakness of the normal equations lies entirely in forming $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, not in factoring it.

```
import numpy as np
import statsmodels.api as sm

def cholesky_upper(A):
    """Upper triangular R with positive diagonal and R'R = A (A positive definite)."""
    A = np.array(A, dtype=float)
    p = A.shape[0]
    R = np.zeros_like(A)
    for j in range(p):
        d = A[j, j] - R[:j, j] @ R[:j, j]
        if d <= 0:
            raise np.linalg.LinAlgError(f"not positive definite at step {j + 1}")
        R[j, j] = np.sqrt(d)
        R[j, j + 1:] = (A[j, j + 1:] - R[:j, j] @ R[:j, j + 1:]) / R[j, j]
    return R

def solve_upper(R, b):
    """Back substitution for R x = b, R upper triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b) - 1, -1, -1):
        x[i] = (b[i] - R[i, i + 1:] @ x[i + 1:]) / R[i, i]
    return x

def solve_lower(L, b):
    """Forward substitution for L x = b, L lower triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b)):
        x[i] = (b[i] - L[i, :i] @ x[:i]) / L[i, i]
    return x
```

[Run this code online](#)

10.2.2 Everything from one factorization

Solving the normal equations $\mathbf{R}^\top \mathbf{R} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ takes two triangular solves: $\mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}$ for $\mathbf{z}$, then $\mathbf{R} \mathbf{b} = \mathbf{z}$. Each costs p^2 flops. It is neater to factor the cross-product matrix of the augmented matrix $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. The factor then contains the solution of the first triangular system and the residual sum of squares.

Proposition 10.2.2. The augmented factorization

Let $\mathbf{X}$ be $n \times p$ of full column rank, suppose $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, and let

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{X}^\top \mathbf{y} \\ \mathbf{y}^\top \mathbf{X} & \mathbf{y}^\top \mathbf{y} \end{pmatrix} = \mathbf{T}^\top \mathbf{T}, \quad \mathbf{T} = \begin{pmatrix} \mathbf{R} & \mathbf{z} \\ \mathbf{0}^\top & d \end{pmatrix},$$

be the Cholesky factorization (Theorem 10.2.1) of the cross-product matrix of $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. Then:

- (a) $\mathbf{R}$ is the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$, the least squares estimate solves $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{z}$, and $d^2 = \text{SSE} = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2$;
- (b) for each $k \leq p$, the residual sum of squares of the regression of $\mathbf{y}$ on the first k columns of $\mathbf{X}$ is $d^2 + \sum_{j>k} z_j^2$. The drop in residual sum of squares when column k joins columns $1, \dots, k-1$ is z_k^2 ;
- (c) $\det(\mathbf{X}^\top \mathbf{X}) = \prod_j r_{jj}^2$, $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$, and the leverage of row $\mathbf{x}_{(i)}^\top$ is $h_{ii} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

Proof. $\mathbf{Z}$ has full column rank because $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, so $\mathbf{Z}^\top \mathbf{Z}$ is positive definite and $\mathbf{T}$ exists. Multiplying out $\mathbf{T}^\top \mathbf{T}$ and comparing blocks with $\mathbf{Z}^\top \mathbf{Z}$ gives

$$\mathbf{R}^\top \mathbf{R} = \mathbf{X}^\top \mathbf{X}, \quad \mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}, \quad \mathbf{z}^\top \mathbf{z} + d^2 = \mathbf{y}^\top \mathbf{y}.$$

The first identity and uniqueness make $\mathbf{R}$ the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$. The second gives $\mathbf{R}^\top \mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{R}^\top \mathbf{z}$ for the solution of the normal equations, so $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{z}$.

For (b), and for the rest of (a), let $\mathbf{Q} = \mathbf{Z}\mathbf{T}^{-1}$. Then $\mathbf{Q}^\top \mathbf{Q} = \mathbf{T}^{-\top} \mathbf{Z}^\top \mathbf{Z} \mathbf{T}^{-1} = \mathbf{I}$, so the columns $\mathbf{q}_1, \dots, \mathbf{q}_{p+1}$ of $\mathbf{Q}$ are orthonormal, and $\mathbf{Z} = \mathbf{Q}\mathbf{T}$. Because $\mathbf{T}$ is upper triangular with nonzero diagonal, the first k columns of $\mathbf{X}$ span the same subspace as $\mathbf{q}_1, \dots, \mathbf{q}_k$. The last column of $\mathbf{Z} = \mathbf{Q}\mathbf{T}$ reads $\mathbf{y} = \sum_{j \leq p} z_j \mathbf{q}_j + d \mathbf{q}_{p+1}$. By Proposition 6.3.2 the projection of $\mathbf{y}$ onto the span of the first k columns is $\sum_{j \leq k} z_j \mathbf{q}_j$, and the residual sum of squares is $\|\mathbf{y}\|^2 - \sum_{j \leq k} z_j^2 = d^2 + \sum_{j>k} z_j^2$. With $k = p$ this is d^2 , which completes (a). Taking differences gives the rest of (b).

- (c) The determinant of a triangular matrix is the product of its diagonal. The inverse formula follows from $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$. Finally $h_{ii} = \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = \mathbf{x}_{(i)}^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{x}_{(i)} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

■

Part (b) says that the squares $z_1^2, \dots, z_p^2$ are the *sequential sums of squares* of the regressors in the order of the columns (Definition 9.3.1). A single factorization gives the whole nested sequence of fits. This is the content of Theorem 6.5.1, now in coordinates. Part (c) costs about np^2 flops for all n leverages, each a triangular solve of p^2 flops.

Example. Murder rates by the augmented factorization

For the state data of Example (Poverty and murder rates) ($n = 50$, with an intercept and the percentages in poverty, in single-parent households and urban, so $p = 4$), the augmented factorization (Proposition 10.2.2) gives the coefficients $-9.0946, 0.2538, 0.3980$ and 0.0046 , and the corner entry $d = 10.0976$, so $\text{SSE} = d^2 = 101.96$. The condition number of $\mathbf{X}$ is only 509, and the solution agrees with a QR solution to a relative difference of 2×10^{-14} . The largest leverage is 0.260, and $\log_{10} \det(\mathbf{X}^\top \mathbf{X}) = 11.14$. The script also checks the standard errors $s \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|$ against statsmodels.

```
data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
```

[Run this code online](#)

```
Z = np.column_stack([X, y])           # augmented matrix [X, y]
S = Z.T @ Z                           # (p+1) x (p+1) cross-product matrix
T = cholesky_upper(S)                 # T = [[R, z], [0, d]]
R, z, d = T[:p, :p], T[:p, p], T[p, p]

beta = solve_upper(R, z)              # R b = z
sse = d ** 2                          # the last diagonal entry is sqrt(SSE)
K = solve_upper(R, np.eye(p))         # R^{-1}
XtX_inv = K @ K.T                     # (X'X)^{-1} = R^{-1} R^{-T}
h = np.array([np.sum(solve_lower(R.T, x) ** 2) for x in X]) # leverages
print("coefficients:", np.round(beta, 4))
print(f"SSE = {sse:.4f}, d = {d:.4f}")
```

[Run this code online](#)

10.2.3 When the normal equations are good enough

Proposition 10.1.3 and Eq. 10.1.2 predict a relative error of order $\kappa^2 u$ in $\hat{\beta}$ from the normal equations, where, as Section 10.5 shows, κ is the condition number of $\mathbf{X}$ after its columns are scaled to equal length. With $\kappa \leq 10^4$ about eight digits survive, more than any coefficient is known statistically. The normal equations are therefore sound for well-conditioned designs, when only cross-products are available, and inside iterations such as the reweighted least squares of Chapter 34 (Theorem 34.4.1). Nearly collinear observational data are where the κ^2 loss bites.

10.2.4 The sweep operator

Before QR became standard, regression programs worked on the augmented cross-product matrix with an operation that adds one variable at a time to the fit. It is still used for fitting many subsets of variables.

Definition 10.2.1. Sweep

Let $\mathbf{A}$ be symmetric with $a_{kk} \neq 0$. **Sweeping** $\mathbf{A}$ on pivot k produces the symmetric matrix $\mathbf{B}$ with

$$b_{kk} = -\frac{1}{a_{kk}}, \quad b_{ik} = b_{ki} = \frac{a_{ik}}{a_{kk}}, \quad b_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

Proposition 10.2.3. What sweeping computes

Let $\mathbf{S} = \mathbf{Z}^\top \mathbf{Z}$ be the cross-product matrix of $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_m]$. Let K be a set of column indices such that $\mathbf{Z}_K$ has full column rank, and let L be the remaining indices. Sweeping $\mathbf{S}$ on the pivots in K , one at a time and in any order, meets only positive pivots and produces the matrix $\mathbf{W}$ with blocks

$$\mathbf{W}_{KK} = -(\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}, \quad \mathbf{W}_{KL} = \mathbf{W}_{LK}^\top = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1} \mathbf{Z}_K^\top \mathbf{Z}_L, \quad \mathbf{W}_{LL} = \mathbf{Z}_L^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{Z}_L,$$

where $\mathbf{M}_K$ is the projection onto $\mathcal{C}(\mathbf{Z}_K)$. Column $j \in L$ of $\mathbf{W}_{KL}$ holds the least squares coefficients of $\mathbf{z}_j$ regressed on $\mathbf{Z}_K$, and $\mathbf{W}_{LL}$ holds the residual cross-products. With $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ and K the columns of $\mathbf{X}$, the last column of $\mathbf{W}$ contains $\hat{\beta}$ and, in its corner, the residual sum of squares.

Proof. We induct on the size of K ; the case $K = \emptyset$ is $\mathbf{W} = \mathbf{S}$. Add a pivot $m \in L$ with $\mathbf{Z}_{K \cup \{m\}}$ of full column rank, and write $\mathbf{G} = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}$, $\mathbf{B}_j = \mathbf{G} \mathbf{Z}_K^\top \mathbf{z}_j$ and $\mathbf{w} = (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m$. The pivot is $\mathbf{z}_m^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m = \|\mathbf{w}\|^2 > 0$. For $i, j \in L \setminus \{m\}$ the new entries are $\mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_K - \mathbf{w} \mathbf{w}^\top / \|\mathbf{w}\|^2) \mathbf{z}_j = \mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_{K \cup \{m\}}) \mathbf{z}_j$ by Lemma 6.6.1. The new entries $\mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in row m , and $\mathbf{B}_j - \mathbf{B}_m \mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in the rows of K , are the coefficients of $\mathbf{z}_j$ on $[\mathbf{Z}_K, \mathbf{z}_m]$ by Theorem 6.6.2. The partitioned inverse (Theorem 1.6.4), with Schur complement $\|\mathbf{w}\|^2$, shows that $(\mathbf{Z}_{K \cup \{m\}}^\top \mathbf{Z}_{K \cup \{m\}})^{-1}$ has blocks $\mathbf{G} + \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $-\mathbf{B}_m / \|\mathbf{w}\|^2$ and $1 / \|\mathbf{w}\|^2$, whose negatives are what the sweep produces: $-\mathbf{G} - \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $\mathbf{B}_m / \|\mathbf{w}\|^2$ and $-1 / \|\mathbf{w}\|^2$. The result depends only on the set K , so the order of the sweeps does not matter. ■

Sweeping a pivot costs about $(p+1)^2$ flops, so a new variable enters a fitted model for $O(p^2)$ work. The price is the same as for the normal equations: the sweep works with cross-products and inherits their κ^2 sensitivity. For the state data, sweeping all four columns reproduces the coefficients and SSE of [Example \(Murder rates by the augmented factorization\)](#). Sweeping only the intercept and the single-parent column puts the SSE of that two-column model, 121.92, in the corner.

```
def sweep(A, k):
    """Sweep the symmetric matrix A on pivot k."""
    A = np.array(A, dtype=float)
    a = A[k, k]
    B = A - np.outer(A[:, k], A[k, :]) / a
    B[k, :] = A[k, :] / a
    B[:, k] = A[:, k] / a
    B[k, k] = -1 / a
    return B

W = S.copy()
for k in range(p):
    W = sweep(W, k)
print("coefficients from the sweep:", np.round(W[:p, p], 4))
print(f"SSE from the sweep: {W[p, p]:.4f}")
```

[Run this code online](#)

10.2.5 Forming cross-products: the centring problem

With an intercept in the model, the cross-product matrix is usually computed in centred form, $\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$, which by Theorem 6.7.1 changes the parameterization but not the fit. How it is computed matters a great deal. The “textbook” formula $\sum_i x_i^2 - n\bar{x}^2$ subtracts two large, nearly equal numbers whenever the mean is large compared with the spread. The loss is governed by the ratio

$$\kappa_{\mathbf{x}} = \frac{\|\mathbf{x}\|}{\|\mathbf{x} - \bar{x}\mathbf{1}\|} = \left(1 + \frac{n\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right)^{1/2},$$

which is the condition number of the column, and it is large when the coefficient of variation is small. Three algorithms are in common use:

- *textbook*: accumulate $\sum x_i$ and $\sum x_i^2$ in one pass, then subtract;
- *two-pass*: compute $\bar{x}$ in a first pass, then sum $(x_i - \bar{x})^2$ in a second;
- *updating*: one pass, maintaining the running mean and the running centred sum by the recursions below.

Proposition 10.2.4. Updating means and centred cross-products

For vectors $\mathbf{x}_1, \mathbf{x}_2, \dots$ in $\mathbb{R}^q$ let $\mathbf{m}_k$ be the mean of the first k and $\mathbf{C}_k = \sum_{i \leq k} (\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)^\top$. With $\mathbf{d}_k = \mathbf{x}_k - \mathbf{m}_{k-1}$,

$$\mathbf{m}_k = \mathbf{m}_{k-1} + \frac{\mathbf{d}_k}{k}, \quad \mathbf{C}_k = \mathbf{C}_{k-1} + \frac{k-1}{k} \mathbf{d}_k \mathbf{d}_k^\top.$$

Proof. The first identity is $k\mathbf{m}_k = (k-1)\mathbf{m}_{k-1} + \mathbf{x}_k$. For the second, write $\mathbf{C}_k = \sum_{i \leq k} \mathbf{x}_i \mathbf{x}_i^\top - k\mathbf{m}_k \mathbf{m}_k^\top$ and put $\mathbf{m} = \mathbf{m}_{k-1}$, $\mathbf{d} = \mathbf{d}_k$. Then

$$\mathbf{C}_k - \mathbf{C}_{k-1} = \mathbf{x}_k \mathbf{x}_k^\top - k\mathbf{m}_k \mathbf{m}_k^\top + (k-1)\mathbf{m} \mathbf{m}^\top.$$

Substituting $\mathbf{x}_k = \mathbf{m} + \mathbf{d}$ and $k\mathbf{m}_k \mathbf{m}_k^\top = k\mathbf{m} \mathbf{m}^\top + \mathbf{m} \mathbf{d}^\top + \mathbf{d} \mathbf{m}^\top + \mathbf{d} \mathbf{d}^\top / k$, all the terms in $\mathbf{m}$ cancel and $\mathbf{d} \mathbf{d}^\top (1 - 1/k)$ remains. ■

The update adds a *centred* rank-one term, so it never subtracts large quantities. It is the cross-product analogue of the row updates of [Section 10.6](#). Chan, Golub and LeVeque (1983) give error bounds of roughly $n\kappa_{\mathbf{x}}^2 u$ for the textbook formula, $n\kappa_{\mathbf{x}} u$ for updating and $nu + n^2\kappa_{\mathbf{x}}^2 u^2$ for the two-pass algorithm.

Example. Centring a column with a large mean

A sample of 1000 standard normal values is shifted by $c = 1, 10, \dots, 10^8$, and the centred sum of squares is computed by the three algorithms and compared with its exact value (found in rational arithmetic). At $c = 10^6$, where $\kappa_{\mathbf{x}} = 1.0 \times 10^6$, the relative errors are 8.9×10^{-5} for the textbook formula, 8.1×10^{-12} for updating and 2.7×10^{-17} for two passes. At $c = 10^8$ the textbook formula has relative error 3.1: not a single correct digit, and possibly a negative “sum of squares”. Updating gives 7.4×10^{-10} and two passes 2.3×10^{-17} . [Figure 10.2.1](#) shows the three growth rates $\kappa^2 u$, κu and u .

```
def ss_textbook(x):
    return np.sum(x * x) - len(x) * np.mean(x) ** 2      # one pass, then subtract

def ss_two_pass(x):
    m = np.mean(x)                                         # first pass: the mean
    return np.sum((x - m) ** 2)                           # second pass: deviations

def ss_updating(x):
    m, s = 0.0, 0.0
    for k, xk in enumerate(x, start=1):                  # one pass, no
        # cancellation
        d = xk - m
        m = m + d / k
        s = s + (k - 1) / k * d * d
```

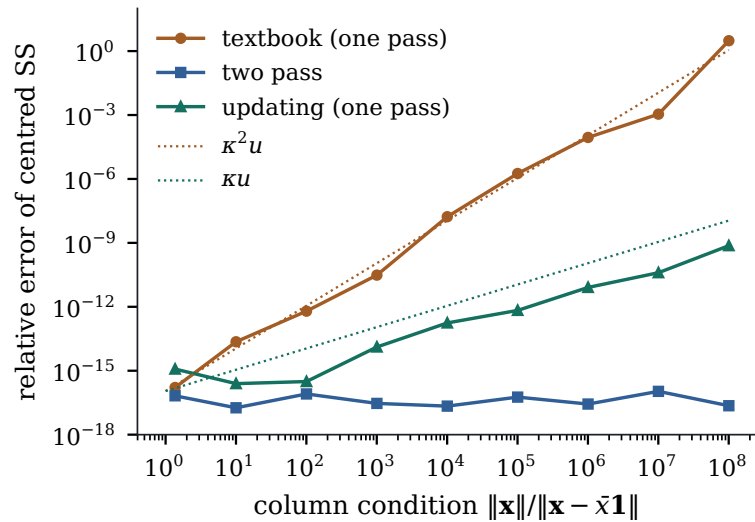


Figure 10.2.1: Relative error of the centred sum of squares of a column of 1000 values with standard deviation one and mean up to 10^8 , against the column condition number $\kappa_{\mathbf{x}}$. The one-pass textbook formula loses accuracy like $\kappa^2 u$, the one-pass updating formula like κu , and the two-pass algorithm not at all.

```
return s
```

[Run this code online](#)

Calendar years, population totals and price indices have large means and small spread. A program that forms cross-products should centre them with two passes (or update), solve for the slopes, and recover the intercept as $\bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_{\text{slopes}}$.

10.2.6 Exercises

10.2.6.1 A. Check your understanding

Exercise. A1

Compute the Cholesky factor of $\mathbf{A} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{pmatrix}$ by Eq. 10.2.1, and use it to solve $\mathbf{A}\mathbf{b} = (2, 14, 6)^\top$ and to find $\det \mathbf{A}$.

Solution. Row 1: $r_{11} = 2$, $r_{12} = 1$, $r_{13} = -1$. Row 2: $r_{22} = \sqrt{10 - 1} = 3$, $r_{23} = (2 - (1)(-1))/3 = 1$. Row 3: $r_{33} = \sqrt{6 - 1 - 1} = 2$. So $\mathbf{R} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $\det \mathbf{A} = (2 \cdot 3 \cdot 2)^2 = 144$. Forward substitution in $\mathbf{R}^\top \mathbf{z} = (2, 14, 6)^\top$ gives $z_1 = 1$, $z_2 = (14 - 1)/3 = 13/3$, $z_3 = (6 + 1 - 13/3)/2 = 4/3$. Back substitution in $\mathbf{R}\mathbf{b} = \mathbf{z}$ gives $b_3 = 2/3$, $b_2 = (13/3 - 2/3)/3 = 11/9$ and $b_1 = (1 - 11/9 + 2/3)/2 = 2/9$. Check: $\mathbf{A}\mathbf{b} = (8/9 + 22/9 - 12/9, 4/9 + 110/9 + 12/9, -4/9 + 22/9 + 36/9) = (2, 14, 6)$.

Exercise. A2

For simple regression, $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, find the Cholesky factor (Theorem 10.2.1) of $\mathbf{X}^\top \mathbf{X}$ and of the augmented matrix, and show that it reproduces the slope S_{xy}/S_{xx} of Theorem 5.2.1. Where does the centring happen?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$ gives $r_{11} = \sqrt{n}$, $r_{12} = \sqrt{n}\bar{x}$ and $r_{22} = (\sum x_i^2 - n\bar{x}^2)^{1/2} = S_{xx}^{1/2}$. The augmented column gives $z_1 = n\bar{y}/\sqrt{n} = \sqrt{n}\bar{y}$ and $z_2 = (\sum x_i y_i - n\bar{x}\bar{y})/S_{xx}^{1/2} = S_{xy}/S_{xx}^{1/2}$. Back substitution gives the slope $b_2 = z_2/r_{22} = S_{xy}/S_{xx}$ and the intercept $b_1 = (z_1 - r_{12}b_2)/r_{11} = \bar{y} - \bar{x}b_2$. The centring happens in the subtraction $a_{22} - r_{12}^2$, which is the textbook formula of Example (Centring a column with a large mean). So the Cholesky factorization of an uncentred cross-product matrix inherits its cancellation.

10.2.6.2 B. Practice

Exercise. B1

Show that the standard error of $\hat{\beta}_j$ is $s \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|$, where $s^2 = \text{SSE}/(n-p)$ (Theorem 5.6.1). Show that all p standard errors cost $p^3/3 + O(p^2)$ flops, and all n leverages $np^2 + O(np)$ flops. Why is computing $\mathbf{M}$ itself a bad idea?

Solution. By Theorem 5.5.1 the estimated variance of $\hat{\beta}_j$ is $s^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$, and $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj} = \mathbf{e}_j^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{e}_j = \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|^2$. The inverse of a triangular matrix is triangular, and column j of $\mathbf{R}^{-1}$ is found by back substitution on its leading $j \times j$ block, at about j^2 flops. The total is $\sum_j j^2 \approx p^3/3$. Each leverage is a triangular solve with $\mathbf{R}^\top$ (p^2 flops) and a squared norm. The projection $\mathbf{M}$ has n^2 entries, which for $n = 10^6$ is too many to store, and only its diagonal is ever needed.

Exercise. B2

Two groups of observations have sizes n_a, n_b , means $\mathbf{m}_a, \mathbf{m}_b$ and centred cross-product matrices $\mathbf{C}_a, \mathbf{C}_b$. Show that the pooled data have centred cross-product matrix

$$\mathbf{C} = \mathbf{C}_a + \mathbf{C}_b + \frac{n_a n_b}{n_a + n_b} (\mathbf{m}_a - \mathbf{m}_b)(\mathbf{m}_a - \mathbf{m}_b)^\top.$$

Show that Proposition 10.2.4 is the special case $n_b = 1$, and explain how the formula allows cross-products to be computed in parallel on separate blocks of data.

10.2.6.3 C. Going deeper

Exercise. C1

Let $\mathbf{A}$ be nonnegative definite of rank r . Show that the Cholesky recursion with *symmetric pivoting*, which at step j moves the largest remaining diagonal entry of the current Schur complement to position j , produces a permutation $\mathbf{\Pi}$ and $\mathbf{\Pi}^\top \mathbf{A} \mathbf{\Pi} = \mathbf{R}^\top \mathbf{R}$ with $\mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ and $\mathbf{R}_{11}$ an $r \times r$ nonsingular upper triangle. (Use Proposition 1.7.5(a).) How is this related to the pivoted QR of Section 10.7?